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TECHNOLOGICAL UNIVERSITY DUBLIN
CITY CAMPUS - GRANGEGORMAN

TU850 - Bachelor of Science (Honours) in
Data Science and Artificial Intelligence

Year 2

SAMPLE PAPER
EXAMINATIONS 2025/26

Generative Artificial Intelligence

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Exam Duration: 2 hours

Instructions: Answer 3 out of 4 Questions

Special Instructions/Handouts: No question is mandatory. Question 1 worth 34 marks, and all other questions are worth 33 marks.

1.

- (a) TensorFlow is a software library developed by the *Google Brain* team, and was initially released in 2015. Explain why TensorFlow is useful in the field of artificial intelligence, and identify four (4) key features of it. (10 marks)

- (b) Explain what the following Python code does (one line at a time). (12 marks)

```
import tensorflow as tf

(x_train, y_train), (x_test, y_test) =
    tf.keras.datasets.imdb.load_data(num_words=10000)

print("Samples 1:", len(x_train))
print("Samples 2:", len(x_test))
```

- (c) Explain what each of the following parameters means: (12 marks)

```
model.fit(epochs=5, batch_size=512, validation_split=0.2)
```

2. (a) Explain the concept of copyright and its role in protecting creative works. Describe how Creative Commons licenses provide alternative ways for creators to share their work, and *discuss two (2) differences* between traditional copyright protection and Creative Commons licensing. (9 marks)
- (b) Artificial intelligence systems are increasingly deployed in critical applications such as finance, healthcare, and infrastructure. Discuss three cybersecurity risks associated with AI systems, and explain how these risks could impact data security, system reliability, or decision-making processes. (12 marks)
- (c) Organizations are increasingly using Generative AI to transform and automate complex business workflows. Explain how Generative AI can support full process reengineering within an organization, and discuss two potential benefits and two challenges associated with integrating GenAI into business processes. (12 marks)

3. NOTE: This question is focused on the *Star Trek: The Next Generation* episode “Measure of a Man” (d: Robert Scheerer, w: Melinda M. Snodgrass), focusing on the legal rights of the android, *Data* (Brent Spiner).
- (a) Captain Picard (Patrick Stewart) asks Commander Bruce Maddox (Brian Brophy), to explain what makes human beings “alive”, and Maddox identifies *three (3) key characteristics of sentience*. Explain each of these characteristics. (12 marks)
 - (b) Commander Maddox believes he can dismantle Data to study him, and then reassemble him later. Maddox acknowledges the risks involved in such a process, but he believes it is worth it for the sake of scientific progress. Discuss *four (4) key ethical responsibilities* that data scientists and AI engineers should have when working with advanced AI systems. (12 marks)
 - (c) When Captain Picard is unsure how to proceed with his case for the defence, he consults with Guinan (Whoopi Goldberg), who suggests an approach he might take, focusing on the potential for slavery in the development of a race of disposable beings. What test could be used to help *determine if a computer system is sentient*, and how is it performed? (9 marks)
- 4.
- (a) Explain the difference between *Supervised Learning* and *Unsupervised Learning*. (8 marks)
 - (b) Pick any two (2) evaluation metrics for text generation, and explain what they are measuring, and how they are measuring it. (10 marks)
 - (c) Generative AI (GenAI) systems can produce highly realistic text communications that closely resemble human-created content. Explain why outputs from Generative AI often feel real or human-like, *identifying five (5) factors* that contribute to this effect. (15 marks)